

15 Graph Theory and Network



15.3 Transition Matrix

Transpose the adjacency matrix and normalize by column to get a transition matrix — applied repeatedly, the state vector converges to a unique steady state

Linear Algebra Made Easy and Visual | Open-source code & drafts: <https://github.com/Visualize-ML>

Core Skills You Will Master

1

Every entry of a transition matrix is a transition probability (a conditional probability)

2

Each column of a transition matrix sums to 1 (a column-stochastic, left stochastic matrix)

3

Powers of the transition matrix combined with a state vector give the weather state after any number of days

4

After many transitions the state vector converges to a fixed probability distribution — the steady state — regardless of the initial state

5

The eigenvector for eigenvalue 1 of the transition matrix, once normalized, is the steady state

From adjacency matrix to transition matrix, steady state, and eigendecomposition

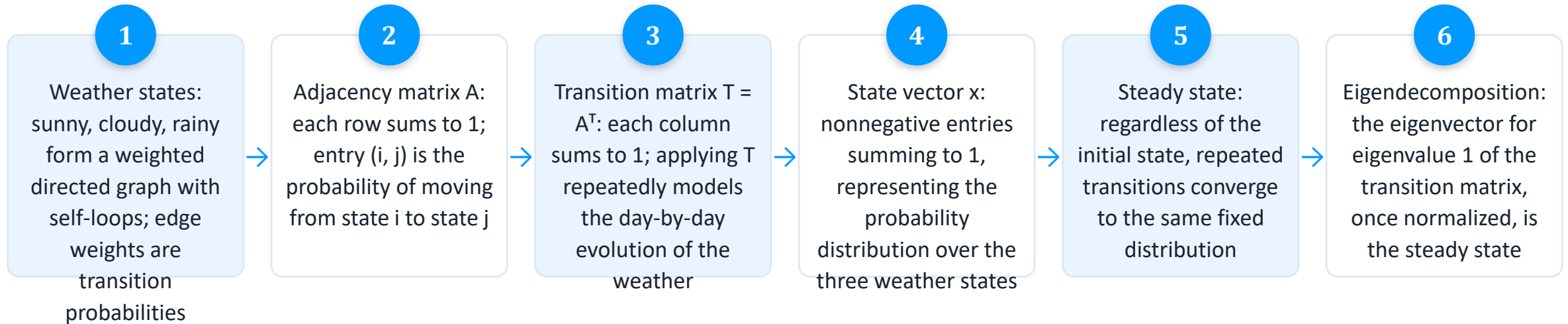


Figure 1 · Three Weather States Transitioning Among Each Other

- Sunny, cloudy, and rainy are 3 nodes; arrows are transition probabilities: from sunny, 70% stays sunny, 25% becomes cloudy, 5% becomes rainy; from cloudy, 45% becomes sunny, 30% stays cloudy, 25% becomes rainy; from rainy, 55% becomes sunny, 30% becomes cloudy, 15% stays rainy
- Every node carries a self-loop — the probability of staying in the same state

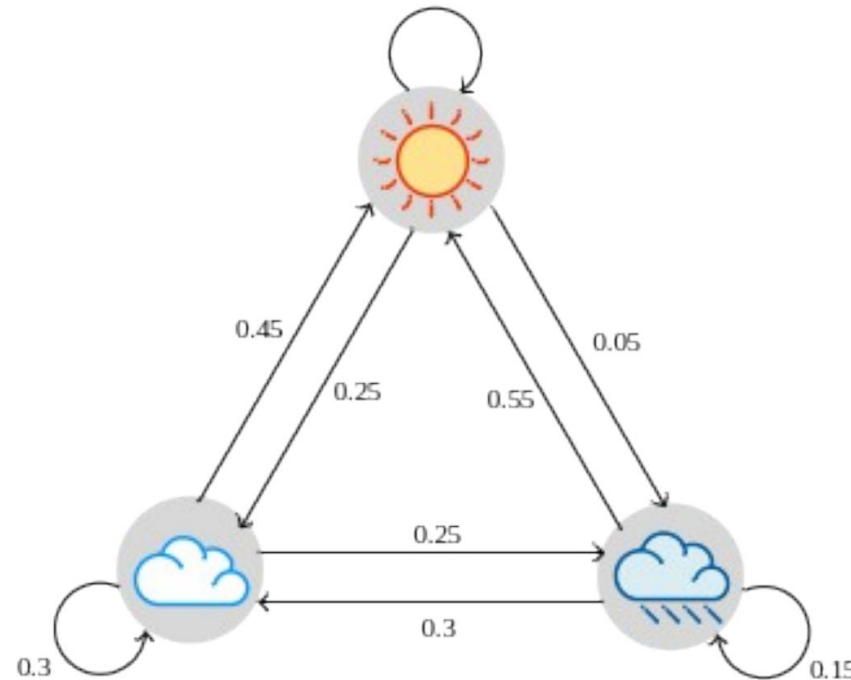


图 1. 三个天气状态相互转换

The Weighted Directed Graph's Adjacency Matrix A Has Rows Summing to 1

$$A = [0.7 \ 0.25 \ 0.05; 0.45 \ 0.3 \ 0.25; 0.55 \ 0.3 \ 0.15]$$

- Entry (i, j) of adjacency matrix A is the probability of moving from state i to state j; each row is the probability distribution over destinations (including staying put) from the current node, so every row sums to 1
- Rows and columns are ordered sunny, cloudy, rainy; row 1, [0.7 0.25 0.05], gives the probabilities of moving from sunny to sunny, cloudy, and rainy the next day

Table 1 · Transition Probabilities Starting From Sunny (Row 1 of A)

- Starting from sunny, the next day is sunny with probability 70%, cloudy with probability 25%, rainy with probability 5% — the three sum to 1

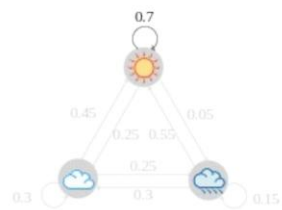
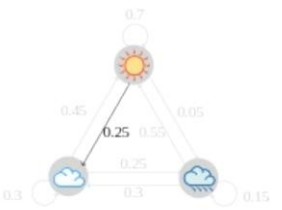
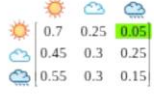
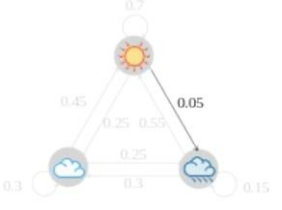
转换	邻接矩阵 A 元素	有向边
晴天 → 晴天 (70%)		
晴天 → 阴天 (25%)		
晴天 → 雨天 (5%)		

Table 1 · Transition Probabilities Starting From Cloudy (Row 2 of A)

- Starting from cloudy, the next day is sunny with probability 45%, cloudy with probability 30%, rainy with probability 25% — the three sum to 1

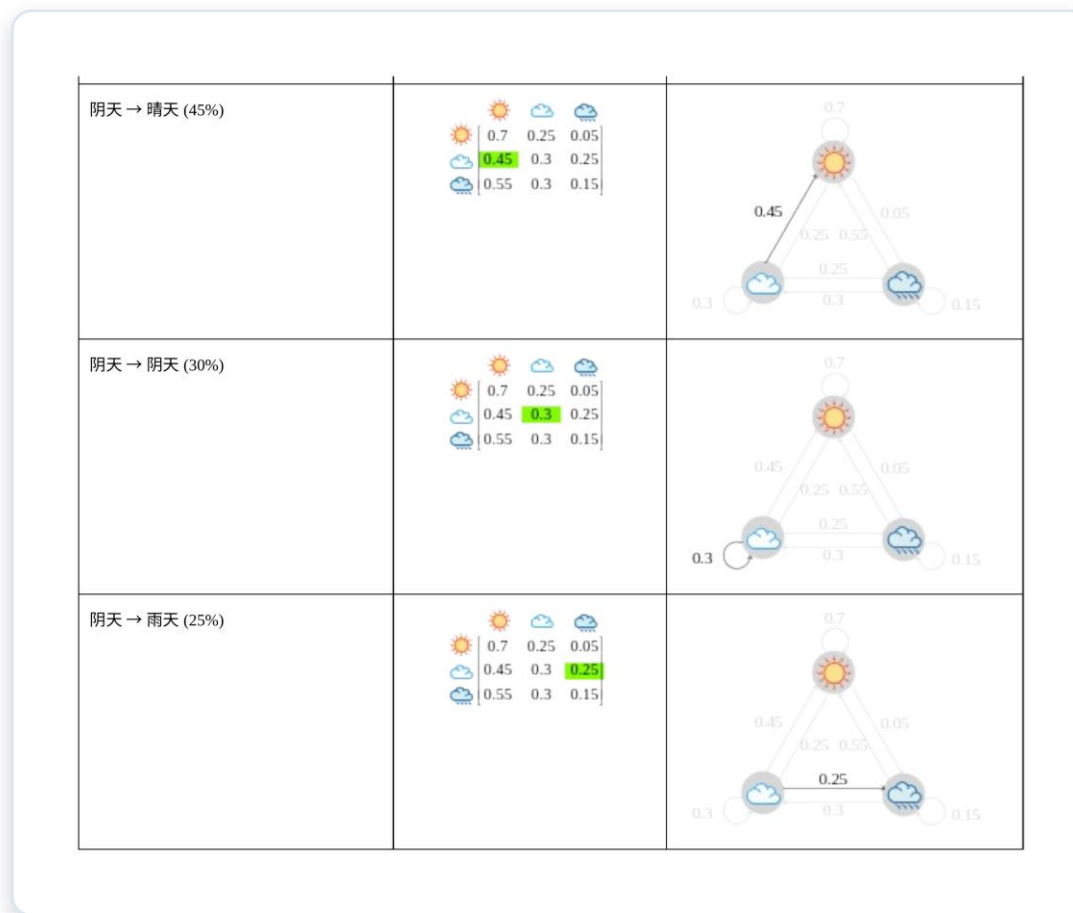
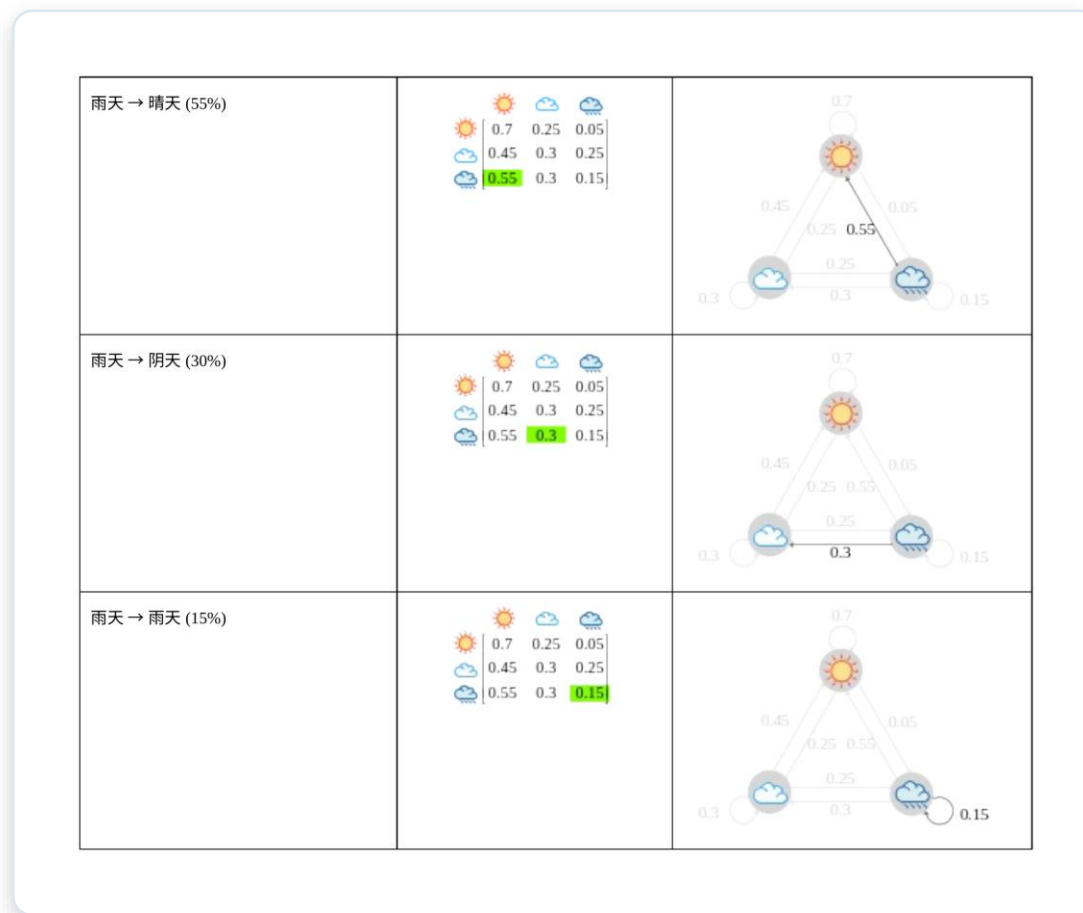


Table 1 · Transition Probabilities Starting From Rainy (Row 3 of A)

- Starting from rainy, the next day is sunny with probability 55%, cloudy with probability 30%, rainy with probability 15% — the three sum to 1



Transposing the Adjacency Matrix Gives Transition Matrix T, With Columns Summing to 1

$$T = A^T = [0.7 \ 0.45 \ 0.55; 0.25 \ 0.3 \ 0.3; 0.05 \ 0.25 \ 0.15]$$

- The transpose A^T of the adjacency matrix A is called the transition matrix (transition matrix, left stochastic matrix, Markov matrix) T ; defined this way, every column of T sums to 1
- State vector x has nonnegative entries summing to 1, e.g. $[P(\text{sunny}) \ P(\text{cloudy}) \ P(\text{rainy})]^T$; applying T to today's state vector gives tomorrow's state vector
- Some references instead use the adjacency matrix A itself (rows summing to 1) as the transition matrix, with the state vector written as a row vector and the multiplication transposed accordingly — this is called a right stochastic matrix; this section uses the more common left stochastic matrix convention

Figure 2 · From Sunny to the Next Day's Weather State

- The current state is certainly sunny, $x_0 = [1 \ 0 \ 0]^T$; $T @ x_0 = [0.7 \ 0.25 \ 0.05]^T$ gives the next day's probability distribution over the three states — equivalent to extracting column 1 of T

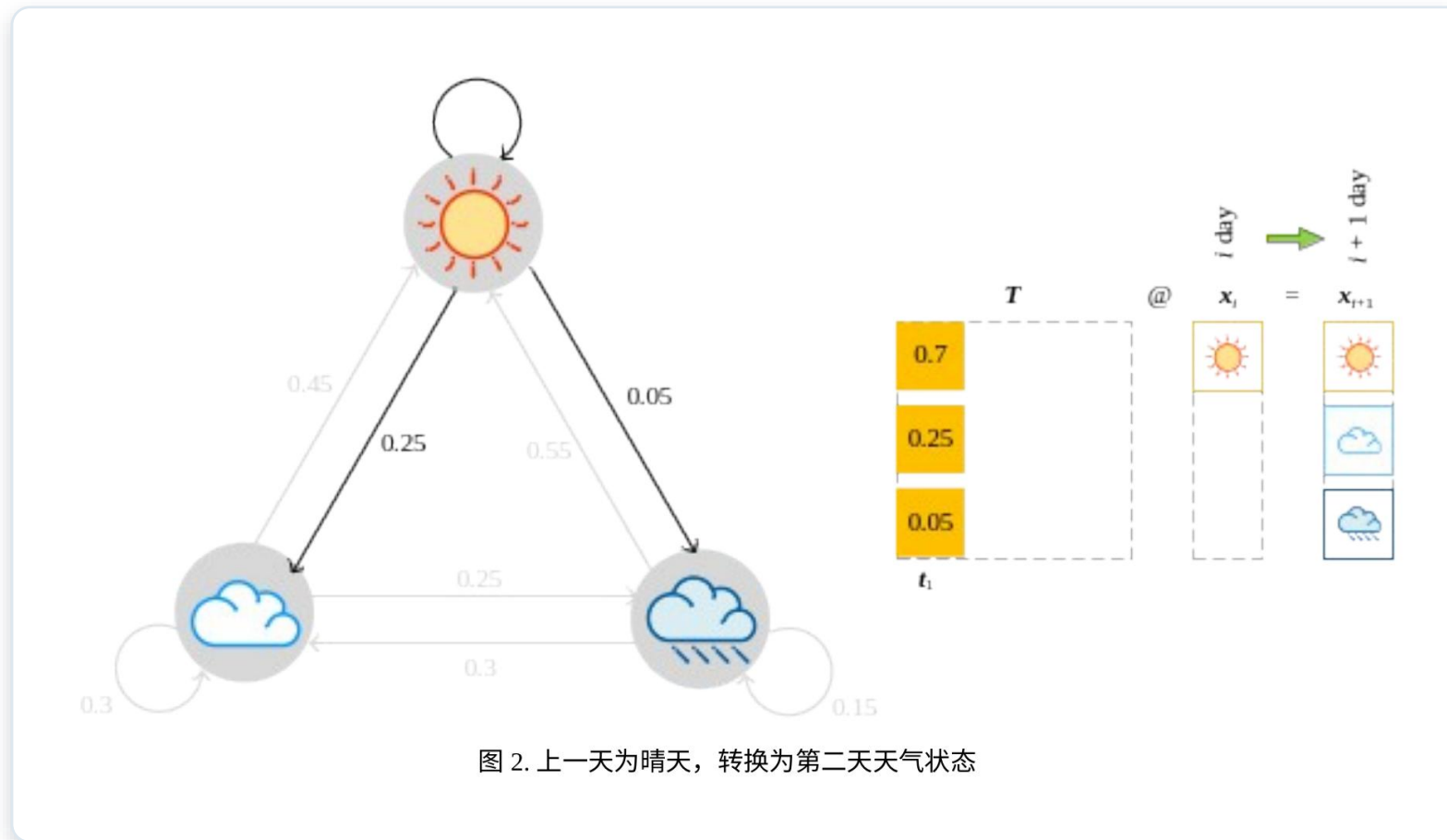


Figure 3 · From Cloudy to the Next Day's Weather State

- The current state is certainly cloudy, $x_0 = [0 \ 1 \ 0]^T$; $T @ x_0 = [0.45 \ 0.3 \ 0.25]^T$ gives the next day's probability distribution — equivalent to extracting column 2 of T

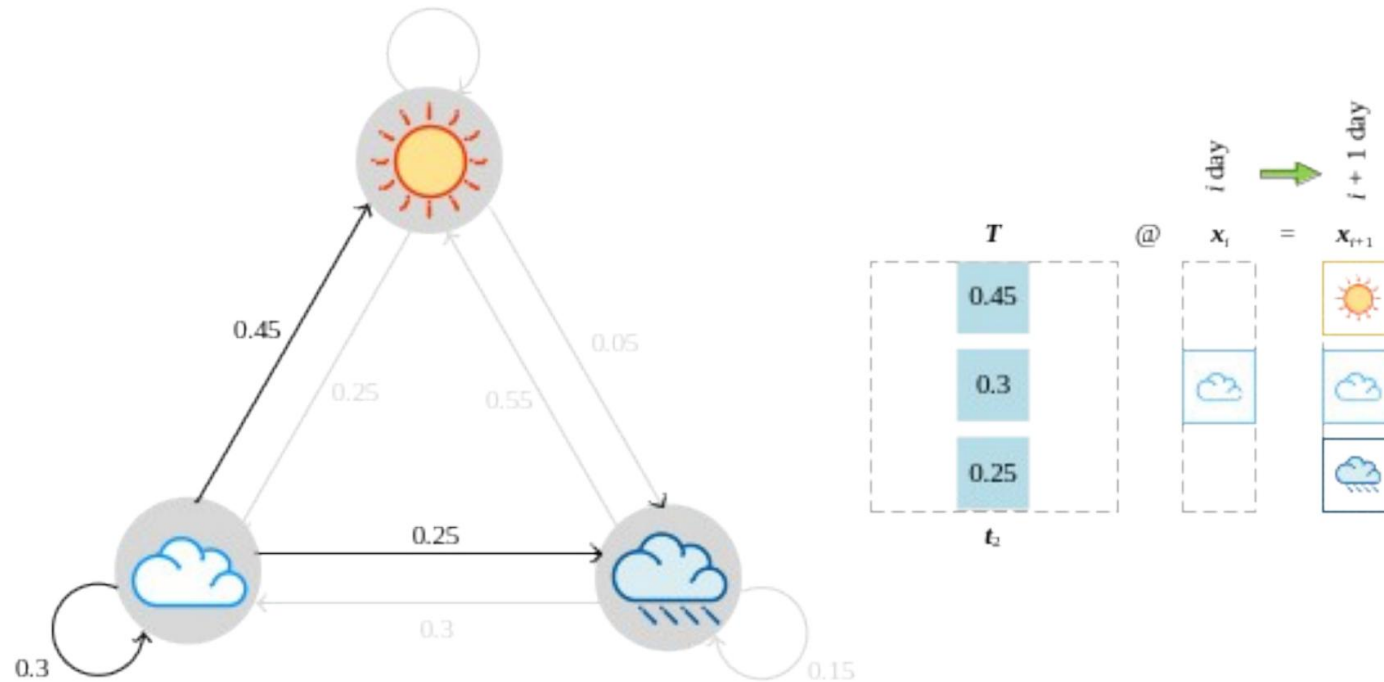


图 3. 上一天为阴天，转换为第二天天气状态

Figure 4 · From Rainy to the Next Day's Weather State

- The current state is certainly rainy, $x_0 = [0 \ 0 \ 1]^T$; $T @ x_0 = [0.55 \ 0.3 \ 0.15]^T$ gives the next day's probability distribution — equivalent to extracting column 3 of T

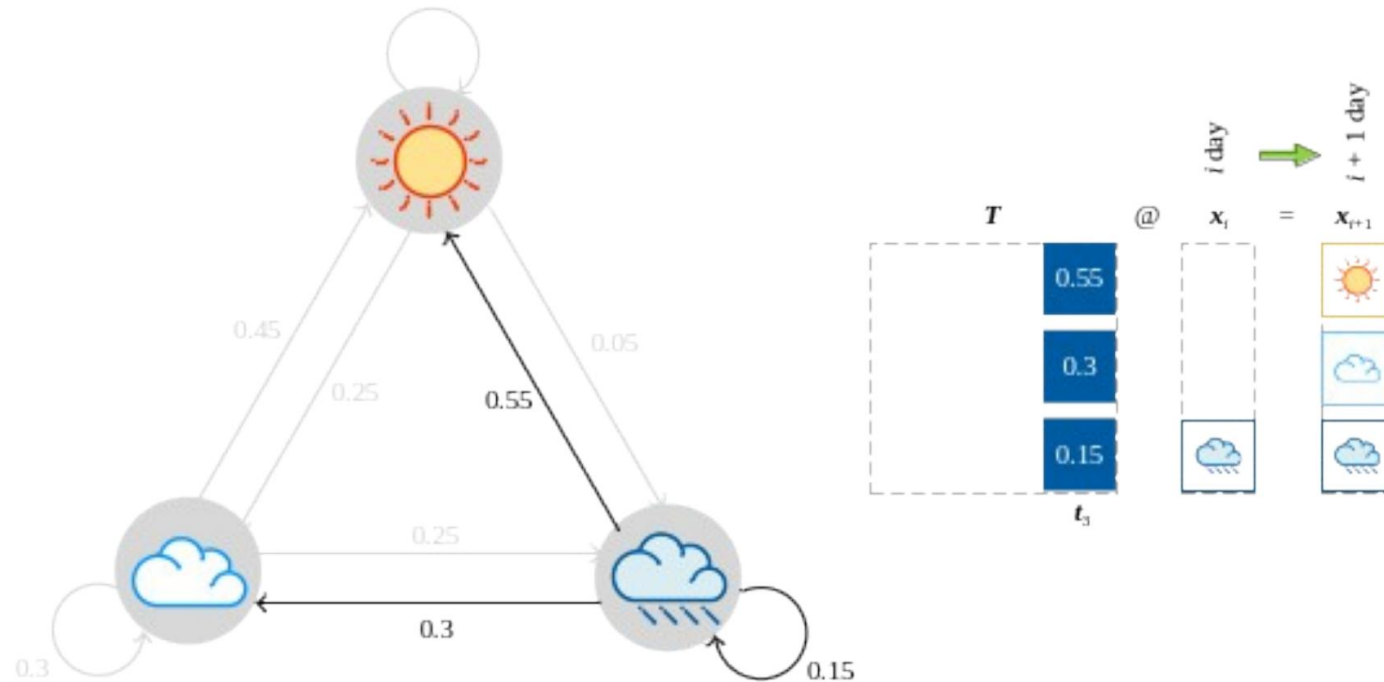


图 4. 上一天为雨天，转换为第二天天气状态

Figure 5 · The Transition Matrix of the Weather-State Directed Graph

- All 9 entries of transition matrix T are shown alongside the weather transitions they represent; applying T repeatedly to the current state vector produces tomorrow's state vector, day after day

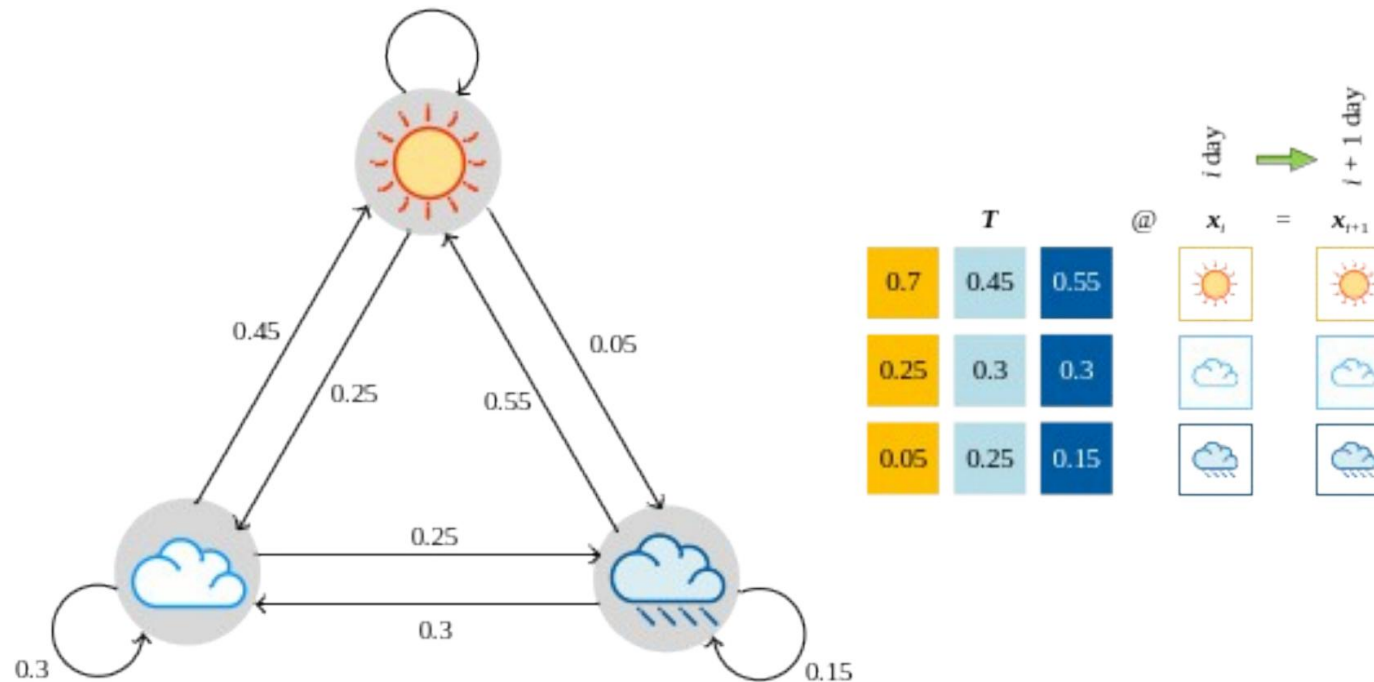


图 5. 天气状态有向图的转移矩阵

Figure 6 · Probability That the Next Day Becomes Sunny

- Row 1 of T , $[0.7 \ 0.45 \ 0.55]$, gives the probability of becoming sunny the next day, starting from sunny, cloudy, or rainy respectively

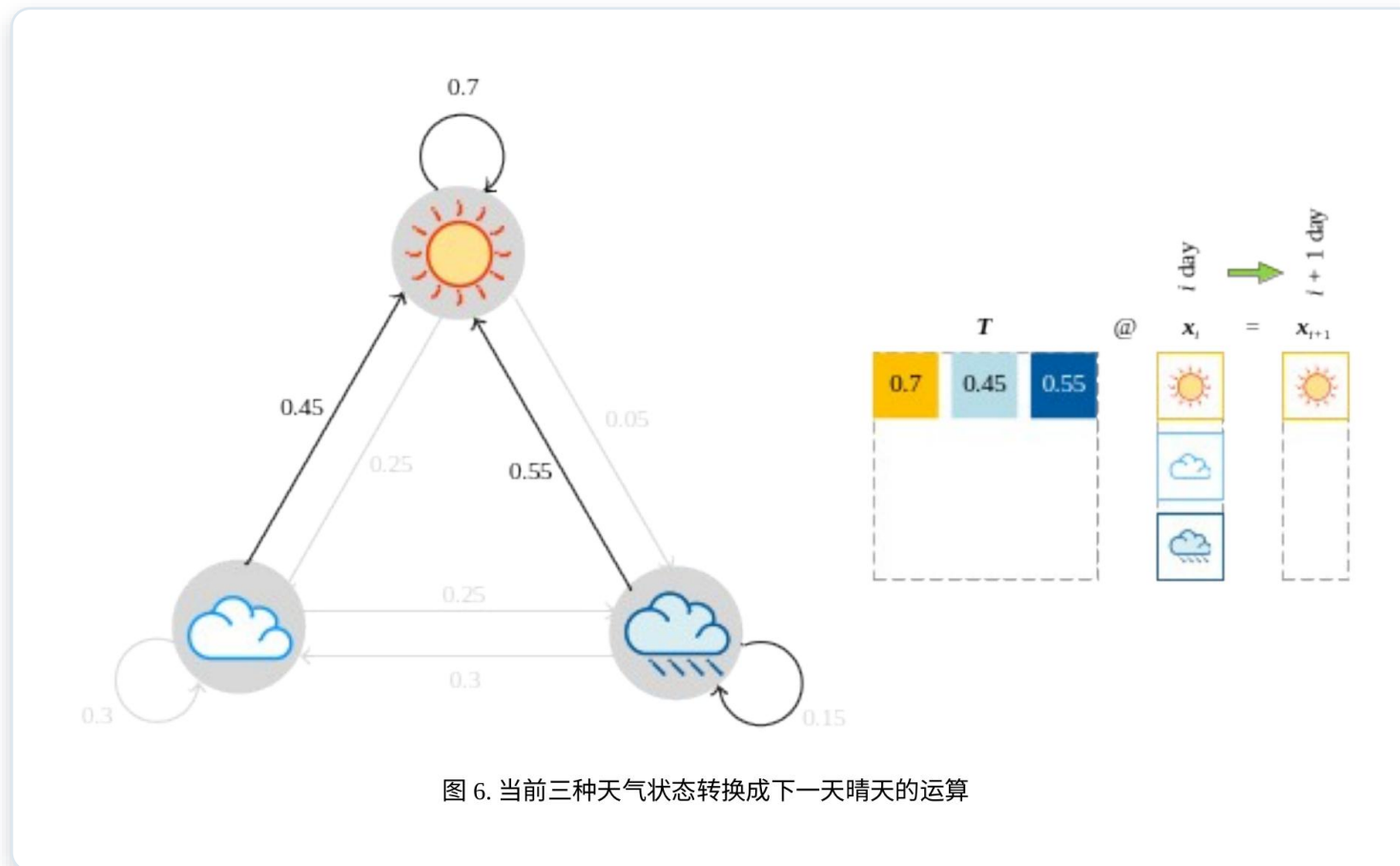


Figure 7 · Probability That the Next Day Becomes Cloudy

- Row 2 of T , $[0.25 \ 0.3 \ 0.3]$, gives the probability of becoming cloudy the next day, starting from sunny, cloudy, or rainy respectively

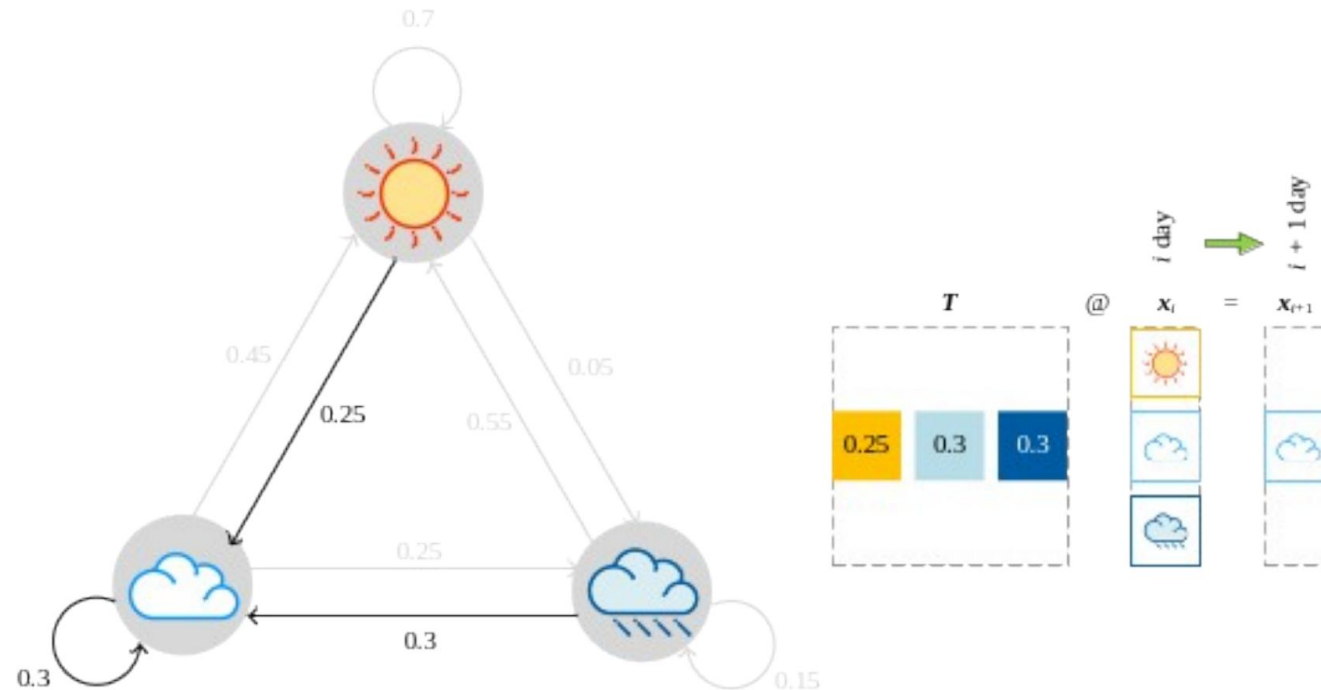


图 7. 当前三种天气状态转换成下一天阴天的运算

Figure 8 · Probability That the Next Day Becomes Rainy

- Row 3 of T , $[0.05 \ 0.25 \ 0.15]$, gives the probability of becoming rainy the next day, starting from sunny, cloudy, or rainy respectively

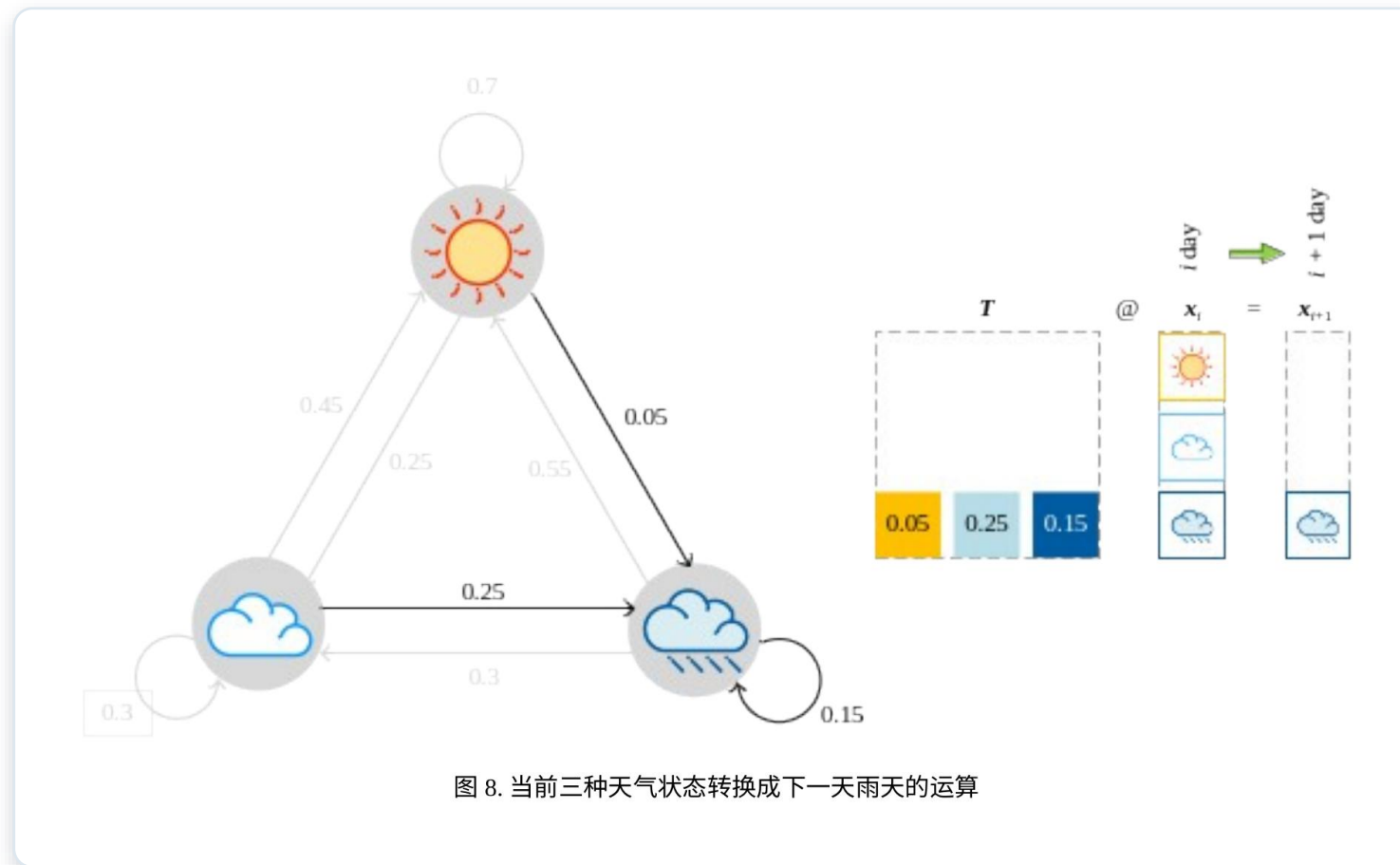


图 8. 当前三种天气状态转换成下一天雨天的运算

Figure 9 · Converging to the Steady State From Sunny

- Starting from $x_0 = [1 \ 0 \ 0]^T$ (sunny); applying T repeatedly, the sunny probability moves from 1.000 to 0.700, 0.630, 0.618, ... and is already very close to $[0.615 \ 0.269 \ 0.115]^T$ after roughly 4–5 days, changing little thereafter



图 9. 从晴天经过转移矩阵变换得到的稳态

Figure 10 · Converging to the Steady State From Cloudy

- Starting from $x_0 = [0 \ 1 \ 0]^T$ (cloudy); applying T repeatedly, the state vector again converges to the same $[0.615 \ 0.269 \ 0.115]^T$ after roughly 4–5 days, matching Figure 9



图 10. 从阴天经过转移矩阵变换得到的稳态

Figure 11 · Converging to the Steady State From Rainy

- Starting from $x_0=[0 \ 0 \ 1]^T$ (rainy); whichever weather state you start from, the sequence converges to the same steady state — much like the ball-passing problem in §15.2, except edge weights are probabilities rather than path counts, and every node now carries a self-loop

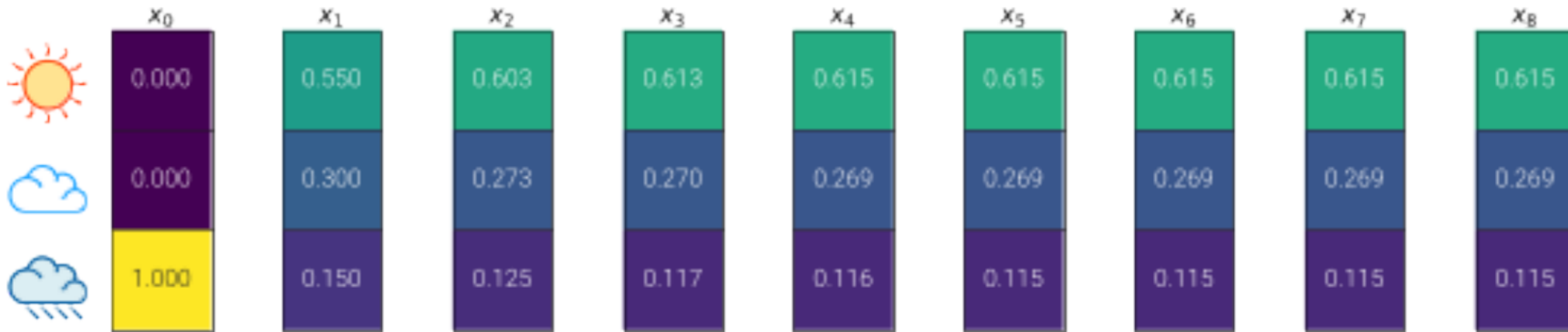


图 11. 从雨天经过转移矩阵变换得到的稳态

The (Normalized) Eigenvector for Eigenvalue 1 Is the Steady State

$$\mathbf{x} = [0.615 \ 0.269 \ 0.115]^T$$

- Eigendecomposing transition matrix T as $T = V\Lambda V^{-1}$, the three eigenvalues are approximately 1, 0.178, and -0.028 (kept to three decimals)
- The eigenvector for eigenvalue 1, once normalized (entries rescaled to sum to 1, not to have unit L2 norm), is exactly the steady state $\mathbf{x}$ that Figures 9–11 converge to
- Vector normalization to unit L2 norm is not the same as normalization to a sum of 1: this section uses the latter, since the entries of a state vector are probabilities and must be nonnegative and sum to 1
- This means that no matter whether today is sunny, cloudy, or rainy, after enough time the weather probabilities always drift toward the same fixed proportions

SUMMARY

Transpose the Adjacency Matrix Into a Transition Matrix; Repeated Application Converges to a Steady State

① Adjacency matrix $\mathbf{A}$ has rows summing to 1; transition matrix $\mathbf{T} = \mathbf{A}^T$ has columns summing to 1

② State vector $\mathbf{x}$ is nonnegative with entries summing to 1; $\mathbf{T}@\mathbf{x}$ gives the next state vector

③ Applying $\mathbf{T}$ repeatedly, regardless of the initial state, the state vector converges to the same steady state

④ The steady state is the eigenvector for eigenvalue 1 of $\mathbf{T}$, normalized so its entries sum to 1

⑤ Unit-L2-norm normalization differs from sum-to-1 normalization; probability distributions use the latter

⑥ Transition matrices and Markov chains are key tools for analyzing how state probabilities evolve over time

Key Terms in This Section

Adjacency matrix A *Matrix of a weighted directed graph; here every row sums to 1*

State vector x *A nonnegative vector with entries summing to 1, a probability distribution*

Eigendecomposition $T=V\Lambda V^{-1}$; *the normalized eigenvector for eigenvalue 1 is the steady state*

Transition matrix T *The transpose A^T of the adjacency matrix; every column sums to 1*

Steady state *The fixed distribution the state vector converges to under repeated transitions*

Markov chain *A random process whose next state depends only on the current state*

FURTHER READING

Further Reading

Review · §15.2 Directed Graphs

Compare the ball-passing problem's adjacency matrix with this section's transition matrix

Further Reading · Data Science: A First Course, Ch. 20

This section is adapted from this book — see the original for more background on Markov chains

Coming Up · §15.4 Laplacian Matrix and Spectral Clustering

Build on the transition matrix to study spectral methods on graphs

Open-Source Code Repository

All code and figures for the book: <https://github.com/Visualize-ML>

Exercises Q1 – Q9

Q1 Compute the eigendecomposition of the following transition matrix, and find its steady state vector: $T = \begin{bmatrix} 0.7 & 0.2 \\ 0.3 & 0.8 \end{bmatrix}$.

Q3 Use NetworkX to build and visualize the directed graph in Figure 1, and compute its adjacency matrix and degree matrix in code.

Q5 Use Python to compute the 2nd and 3rd powers of this section's transition matrix T ; check whether every power still has columns summing to 1.

Q7 Use Python to compute the eigenvector for T 's eigenvalue 1, normalize it so its entries sum to 1, and compare it with this section's steady state vector.

Q9 Starting from sunny, use Python to iterate the weather state vector until the difference between consecutive days is below $1e-6$, and report how many days that takes.

Q2 Given three different initial state vectors $x_0 = \begin{bmatrix} 1 & 0 \end{bmatrix}^T$, $\begin{bmatrix} 0 & 1 \end{bmatrix}^T$, and $\begin{bmatrix} 1/2 & 1/2 \end{bmatrix}^T$, use Q1's transition matrix to iterate 10 times each; compare whether the three converge to the same state vector.

Q4 Verify whether every row of this section's adjacency matrix A sums to 1 and every column of transition matrix T sums to 1, and explain the transpose relationship between them.

Q6 Use Python to eigendecompose this section's 3×3 transition matrix T ; verify that one of its eigenvalues is exactly 1.

Q8 Explain the difference between unit-L2-norm normalization and sum-to-1 normalization; use Python to apply both to the same vector and compare the results.

Exercises Q10 – Q17

Q10 Repeat Q9's iteration starting from cloudy and from rainy; compare whether all three initial states take the same number of days to reach the steady state.

Q12 Explain why, regardless of the initial weather state, the state vector converges to the same steady state vector after enough days.

Q14 Design your own 3×3 transition matrix whose columns sum to 1, and use Python to compute its steady state vector.

Q16 Give examples of other real-world scenarios that could be modeled with transition matrices and Markov chains, such as brand switching or web-page navigation.

Q11 Use Python to plot the three sunny-probability convergence curves from Figures 9–11 against the number of days; check whether they all converge to the same value.

Q13 Use Python to verify that the steady state vector x satisfies $Tx=x$, i.e. x is the eigenvector of T for eigenvalue 1.

Q15 Use Python to verify that the steady state vector of your Q14 matrix has entries summing to 1 and that all entries are nonnegative.

Q17 Write a Python function that checks whether a given square matrix is a valid transition matrix (columns summing to 1).

Exercises Q18 – Q25

Q18 Explain the transpose relationship between adjacency matrix A and transition matrix T , and why applying T to a state vector models probability evolving over time.

Q20 Referencing §15.2's ball-passing problem, explain the similarities and differences between a transition matrix and the ball-passing problem's adjacency matrix (hint: self-loops, and whether edge weights are probabilities).

Q22 Use Python to verify whether a transition matrix can still have a steady state when it contains self-loops (nonzero diagonal entries).

Q24 Explain why the eigenvalue corresponding to a transition matrix's steady state vector must be 1, and not any other value.

Q19 Use Python to compute the 10th power T^{10} of this section's transition matrix T ; check whether every column is already very close to the steady state vector.

Q21 Use Matplotlib to plot a heatmap of this section's transition matrix T , and compare it against the edge weights in Figure 1's directed graph.

Q23 Use Python to compare, for different initial states, how far the state vector is from the steady state after 5 iterations versus after 20 iterations.

Q25 Preview §15.4 and hypothesize what connection the Laplacian matrix might have to this section's transition matrix, with your reasoning.

10 Prompts to Explore With AI

1

Explain the transpose relationship between the transition matrix and the adjacency matrix, and why every column of a transition matrix sums to 1.

2

Explain what the state vector x represents, and why its entries must be nonnegative and sum to 1.

3

Explain why, regardless of the initial weather state, the state vector converges to the same steady state after enough days.

4

Explain the relationship between the eigenvector for eigenvalue 1 of the transition matrix and the steady state.

5

Explain the difference between unit-norm normalization and sum-to-1 normalization, and why this section normalizes the eigenvector the latter way.

10 Prompts to Explore With AI

6

Give examples of other real-world scenarios that could be modeled with Markov chains and transition matrices.

7

Compare the ball-passing problem's adjacency matrix from §15.2 with this section's weather transition matrix, and explain their structural similarities and differences.

8

Explain why every power of a transition matrix (its square, its cube, and so on) still has columns summing to 1, and what that means probabilistically.

9

Explain why, in the eigendecomposition $T=V\Lambda V^{-1}$, only the eigenvector for eigenvalue 1 is the steady state, and what role the other eigenvalues play.

10

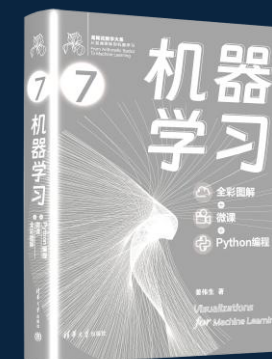
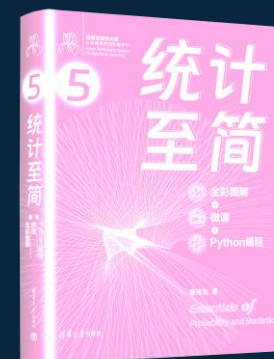
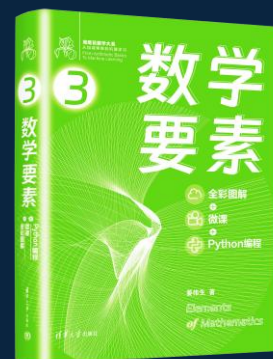
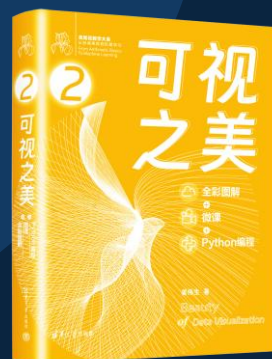
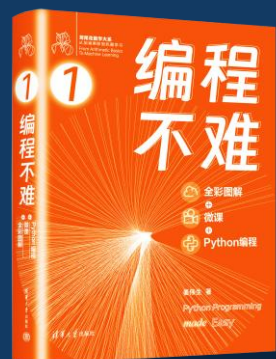
Looking ahead to §15.4's Laplacian matrix and spectral clustering, speculate what connection it might have to this section's transition matrix and Markov chains.

数学

数学基础

线性代数

概率统计



Python编程

数据可视化

工具

数据分析
图和网络

回归、降维
分类、聚类

实践

